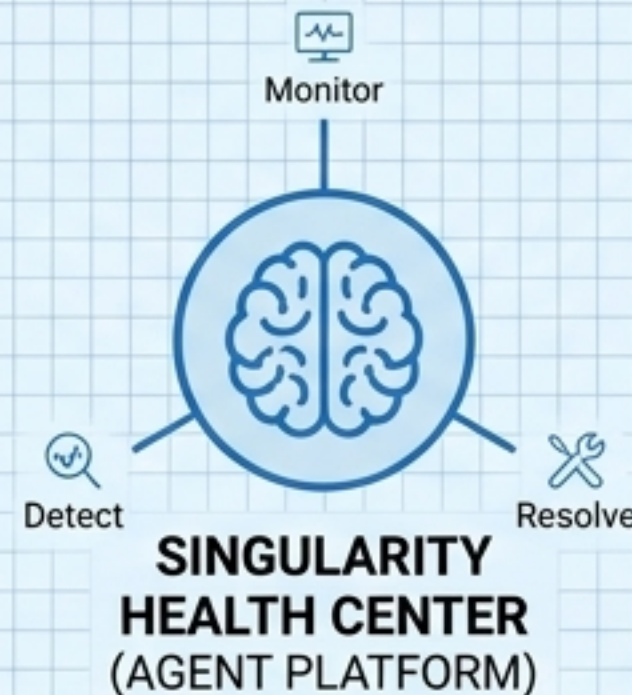


Blueprinting the Singularity Health Center

Delivering a hyperscale agent platform engineered to monitor, detect, and resolve operational health issues across millions of SentinelOne endpoints.

ENGINEERING OBJECTIVES

- **HYPERSCALE MONITORING**
- **PRECISION DETECTION**
- **AUTONOMOUS RESOLUTION**



OPERATIONAL METRICS

- **ENDPOINTS: 10+ MILLION**
- **LATENCY: <5ms (P95)**
- **DETECTION RATE: >99.99%**

Trust Requires Mathematical Promises

We are building an authoritative operational health platform. Missing data is a signal, but it is not proof of endpoint failure. We must mathematically differentiate between endpoint offline, network partition, and pipeline delay.

SCALE



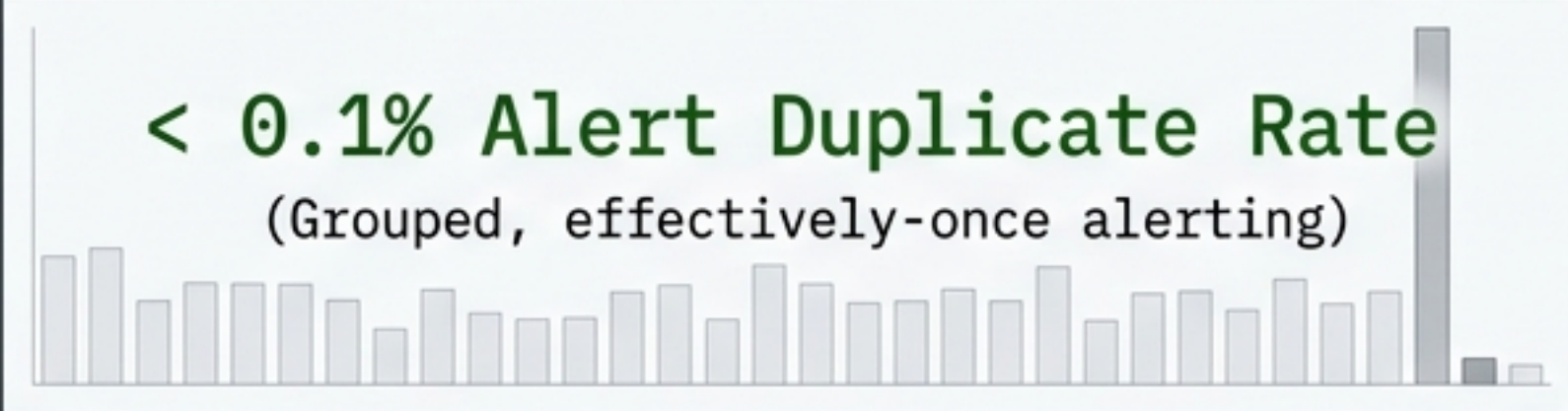
RESILIENCE



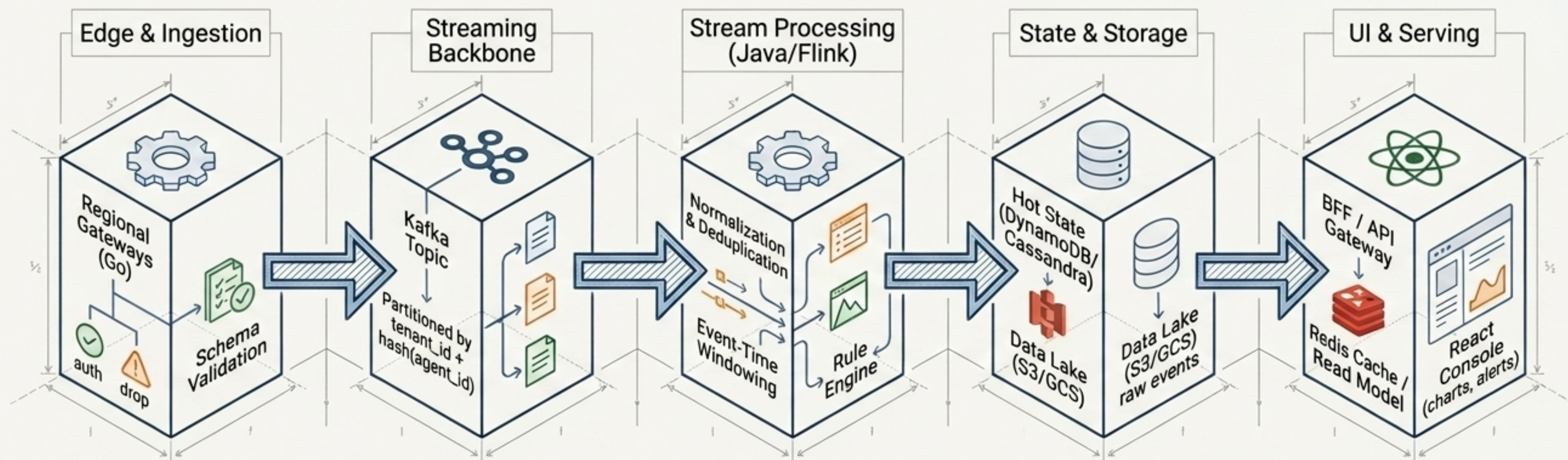
SPEED



PRECISION



End-to-End Streaming Backbone: Final Product System Design & Data Flow



1. Edge & Ingestion

Regional Gateways (Go) terminate auth and drop invalid schemas. Fast ACKs protect endpoints.

2. Streaming Backbone

Partitioned by tenant_id + hash(agent_id) to isolate noisy tenants while preserving order.

3. Processing (Java/Flink)

Normalization & Deduplication
Event-Time Windowing
Rule Engine

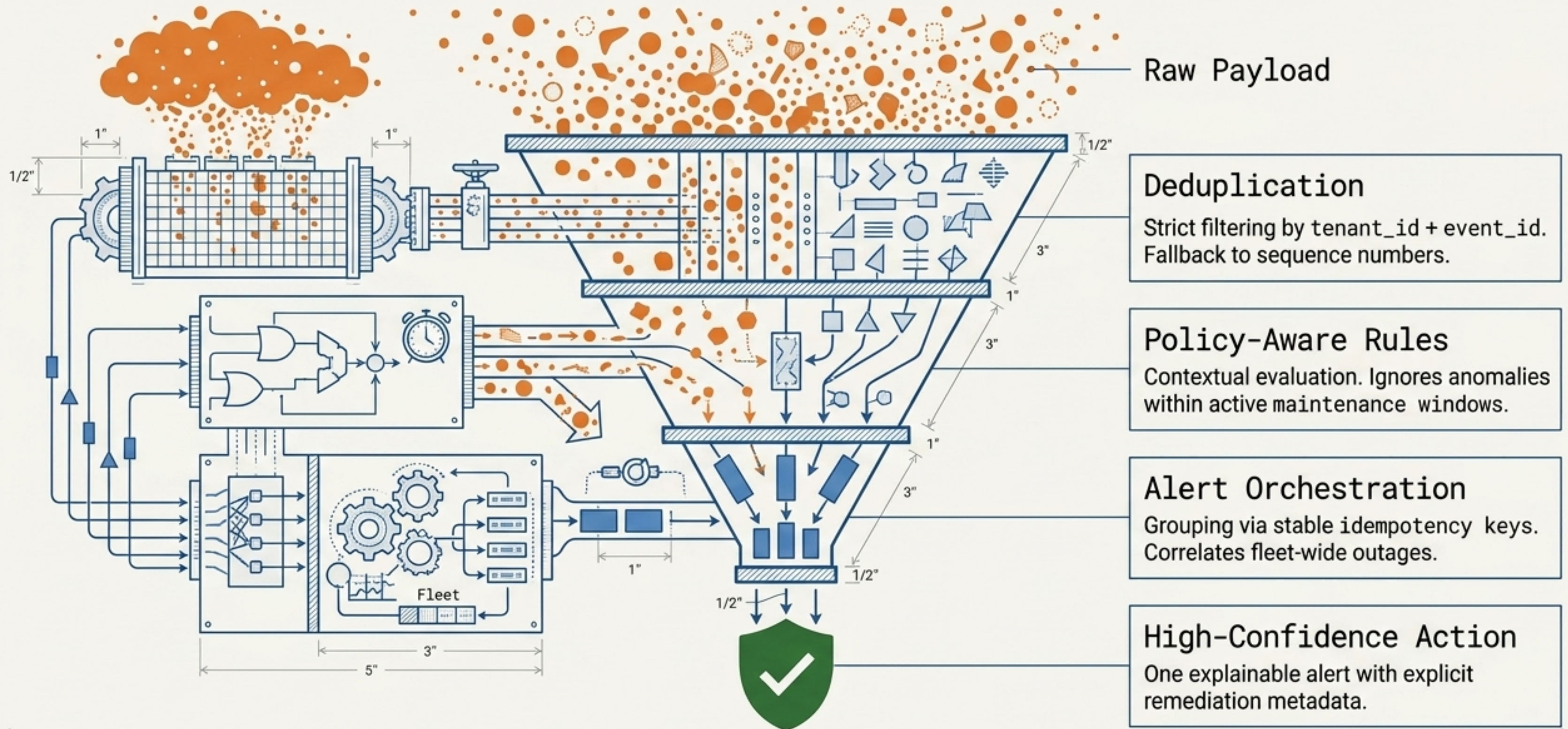
5. Alerting & Action

Alert Orchestration
Action Service (external integrations)

6. UI & Serving

BFF / API Gateway
Redis Cache / Read Model
React Console (charts, alerts)

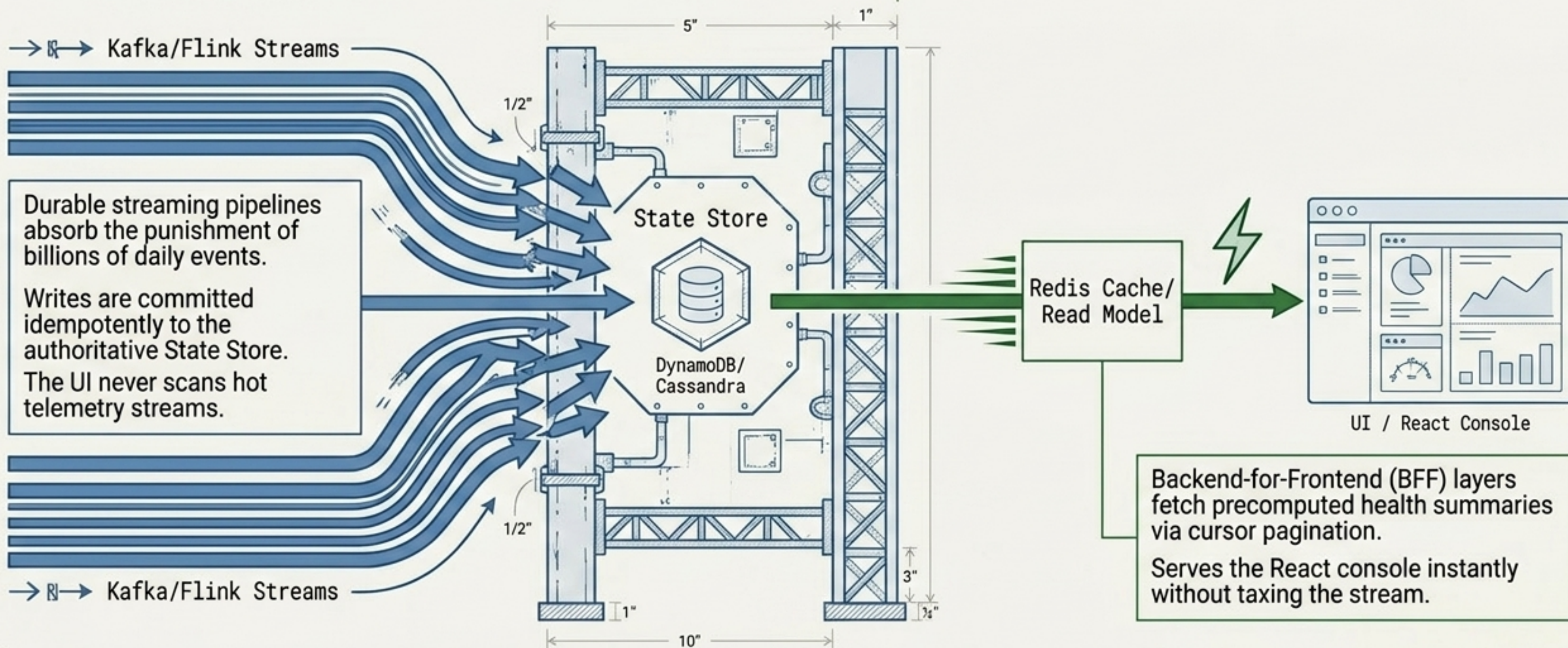
Protecting Customer Trust Over Alerting Volume



Isolating High-Volume Writes From Low-Latency Reads

The Write Path: Massive Throughput

The Read Path: Sub-200ms Access



The Sev-1 Response: Surviving a Silent Stop

Scenario: The Sev-1 Response: a Silent Stop surviving readsann, Kafka/Flink response and ensure that on a consumer denies convrats event, stop proteins the blueprint and promoter tement isned reliability.

1. Consumer Lag Spikes



Consumer lag spikes and consumer's timestamps.

2. Oldest Event Age



Oldest event age stove in tilraes and oldest event compitimes.

3. Synthetic Canaries



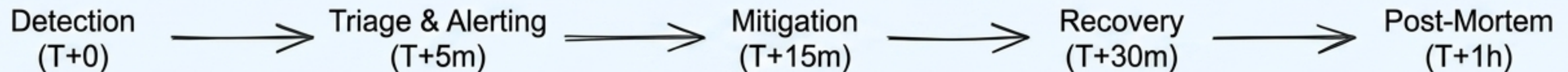
Synthetic canaries when relieving status in actually timestamps.

4. Stale Timestamps



Stale timestamps usen tied to stale timestamps.

Sev-1 Execution Timeline & Runbook

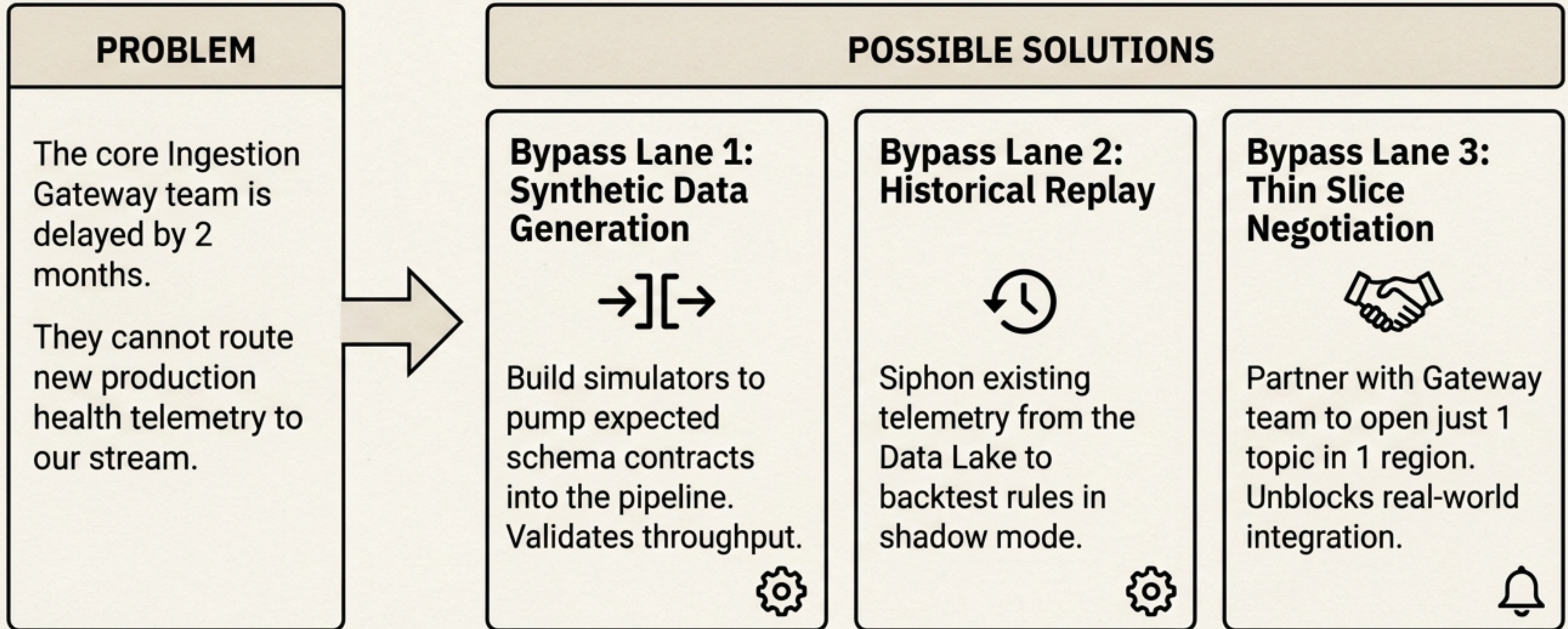


- Runbook:
- | | | |
|--|---|---|
| ☐ Verify Alerts | ☐ Validate UI health | ☐ Communicate status |
| ☐ Engage on-call | ☐ Initiate manual failover | ☐ Conduct RCA. |
| ☐ Check Kafka/Flink metrics | ☐ Monitor manual | |

Pragmatic Phasing: Proving Value Before Proving Scale

Dimension	Minimum Viable Product (MVP)	General Availability (GA)
Primary Goal 	Prove core pipeline & establish observability 	Safely replace legacy health status globally 
Alert Routing 	Shadow mode execution, limited preview tenants 	Default enablement with broad notification fanout 
Rule Scope 	Core 4 rules (Tamper, Disabled, Disk, Heartbeat) 	Fleet-wide groupings, aggregations, and policy drift 
Exit Criteria 	5-min latency achieved, zero UI raw scans 	Sev-1 Game Days passed, legacy systems retired 

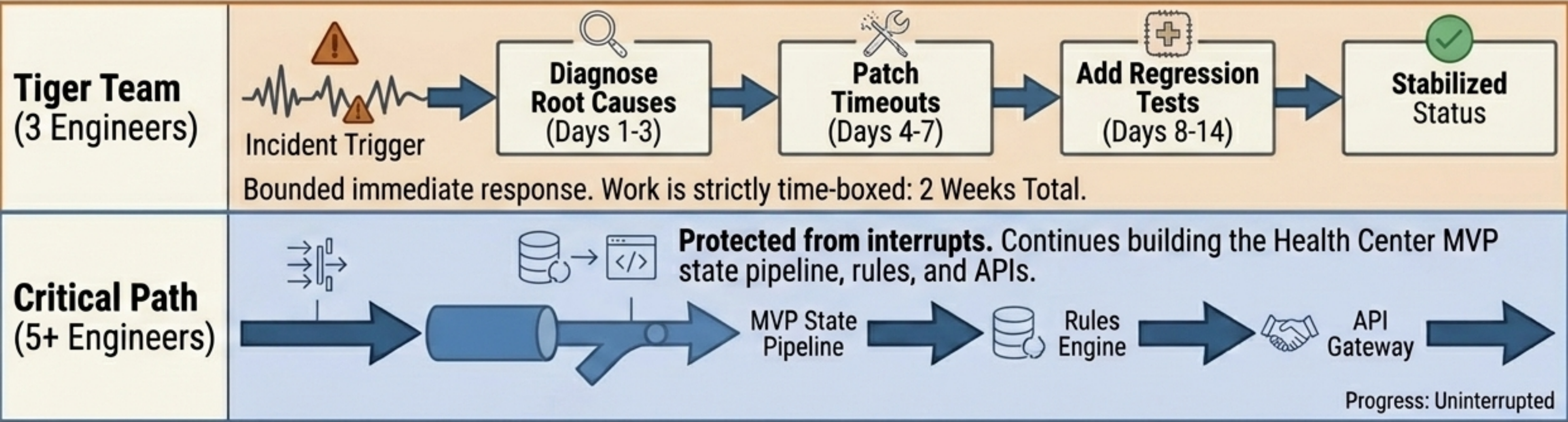
Preserving forward momentum despite external dependency risk



Quarantining the Operational Drain

The Crisis: A 40% spike in customer escalations due to legacy heartbeat microservice showing false 'Offline' statuses.

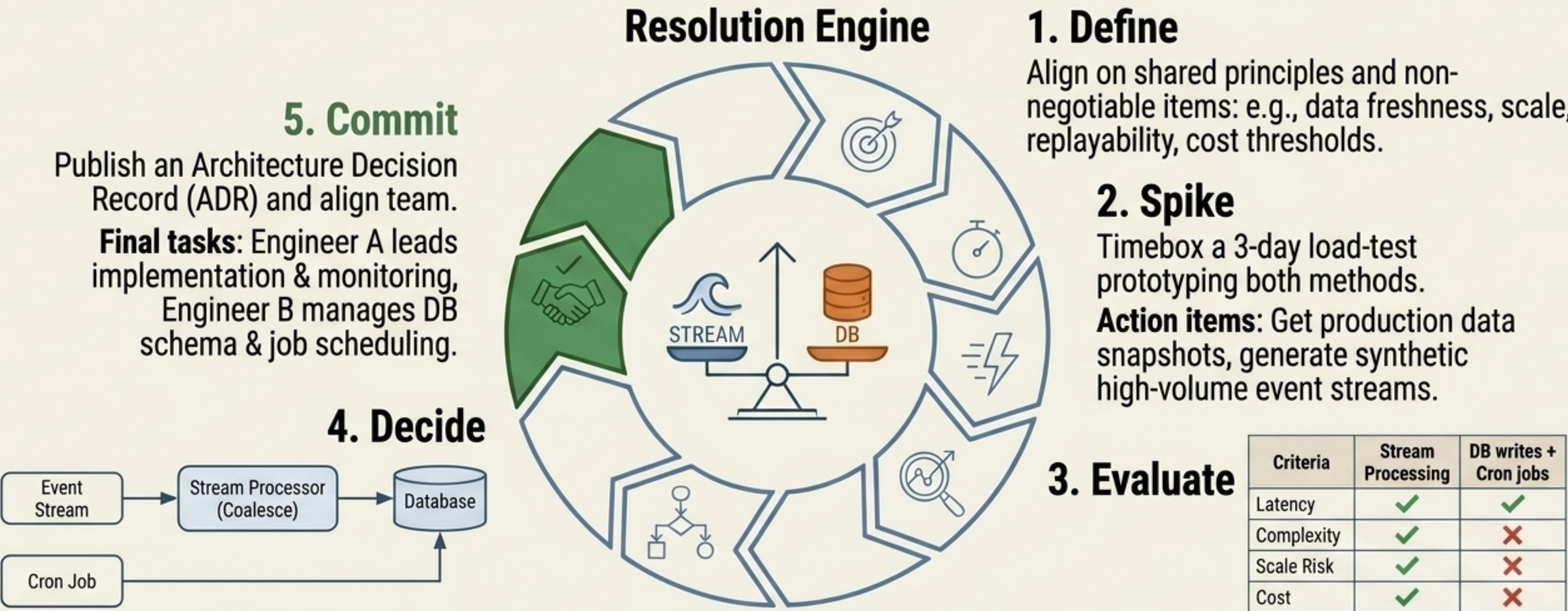
Capacity Allocation Swimlanes



Strategic Outcome: We do not sacrifice current customer trust to ship the future, but we refuse to let legacy fires consume the entire team.

Resolving the Alert Coalescing Conflict

The Conflict: Engineer A argues for Stream Processing (optimizes for 5-min latency, high complexity). Engineer B argues for DB writes + Cron jobs (optimizes for simplicity, risks scale/delays).



Shielding Morale Through Ambiguity

The Challenge: Team morale suffers when engineers feel buffeted by unpredictable legacy issues and undefined work, leading to burnout and attrition.

Full-Stack Team (10 Engineers)

• Backend



• Frontend



• QA



Visible Reality

Maintaining a **transparent roadmap**. By explicitly tracking dependency blocks and legacy stabilization as **planned work**, we eliminate ambient anxiety.

Bounded Interruptions

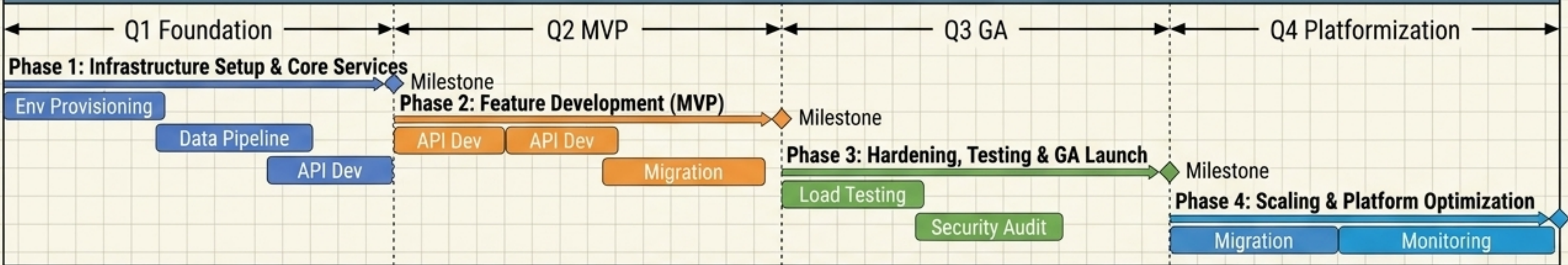
Rotating **incident commanders**. Engineers know exactly when they are the shield (handling the legacy spike) and when they are protected.

Meaningful Agency

Turning operational fires into growth opportunities. Senior engineers drive **ADRs**; QA leads **replay automation**. Progress is celebrated.

Singularity Health Center: The Path Forward - Project Management & Resource Plan

OVERALL PROJECT MANAGEMENT TIMELINE



RESOURCE PLAN (KEY ROLES & FOCUS)

Role	Icon	Q1	Q2	Q3	Q4
Tech Lead		Full	Full	Full	Full
Backend Eng (x3)		Full	Partial	Partial	Full
Frontend Eng (x2)		Full	Partial	Partial	Full
QA Lead		Full	Partial	Partial	Full
DevOps		Full	Full	Full	Full

OPERATING CADENCE

Meeting	Frequency	Purpose / Attendees
Sprint Planning	Bi-weekly	Define scope, estimate tasks. All Eng.
Daily Standup	Daily	Quick sync on progress & blockers. All Eng.
Sprint Review/Demo	Bi-weekly	Showcase completed work. Stakeholders & Eng.
Steering Committee	Monthly	Strategic alignment & budget. Leadership & Tech Lead.